1. Vectors

A vector is an ordered list of numbers. Vectors add component-wise.
MARKER-VEC-P1. The span is all linear combinations.

2. Vector spaces

A vector space is closed under addition and scalar multiplication. Examples R^n and polynomials degree <= n. A basis is a linearly independent spanning set. MARKER-BASIS-P2.

3. Matrix operations

Matrices are rectangular arrays. Matrix multiplication is associative
but NOT commutative. MARKER-COMM-P3. Transpose swaps rows and columns.
Invertible iff determinant non-zero.

4. Determinants

The determinant of a 2x2 matrix [[a,b],[c,d]] is ad - bc. MARKER-DET-P4. Geometrically it scales area/volume. Zero determinant collapses space.

5. Eigenvalues

An eigenvector of A is non-zero v with Av = lambda v. MARKER-EIGENVEC-P5. Eigenvalues are roots of det(A - lambda I) = 0.

6. Diagonalisation

A matrix with n independent eigenvectors factors as A = P D P^-1. MARKER-DIAG-P6. Symmetric matrices have real eigenvalues and orthogonal eigenvectors.